

Arduino Shutter Tester Version 2

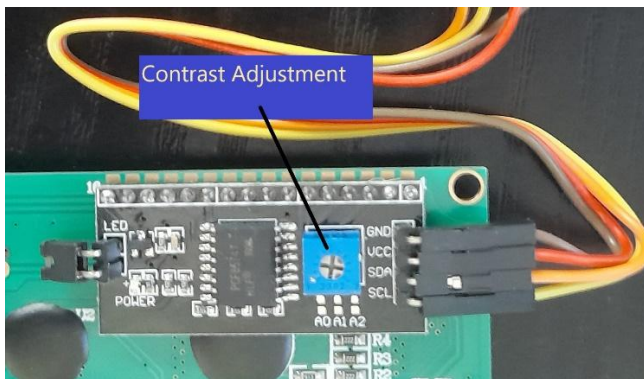
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This document is for firmware 2.0.5 and later. For older firmware, refer to the previous Operating Guide.

First use after loading firmware

Assuming the build has been completed correctly and the firmware loaded, when the Shutter Tester is connected to the computer via the USB cable, The LCD should light and display text.

If the LCD lights, but no text can be seen, try adjusting the contrast screw on the back. As an initial setting, turn fully clockwise and then just a small amount back.



User Authentication.

The first time the Shutter Tester is operated, the LCD and PC screen will say 'Auth Code' and display a number. This number must be sent via private message to Niglyn. The quickest way is via the Photrio forum, link at the top of the page.

A user key will be sent by return and must be entered into the shutter Tester. This is only required once.

To enter the user key, press the **Right Button**. Each press increases the displayed number by 1.

Keeping the button pressed will rapidly increase the displayed number. A quick press will increase the number by 1.

When the supplied user key matches that displayed, press the **Left Button**.

Be careful not to overshoot the number, else you will need to continue to 9999 then keep pressing, as the number wraps round to 0.

Alternately, press the Left Button. The Shutter Tester will restart and you can try again.

Note each incorrect attempt will cause a greater delay before the pass key can be re-entered.

Once entered correctly, the screen will say 'Correct'. After a short while the screen will blank & go dark. After some internal number crunching, the Shutter Tester will spring back into life.

Note Most output, (other than Laser Alignment Utility) is also sent to the PC screen using the Arduino IDE Serial Monitor, as described later in this document.



Splash Screen.

Laser Alignment Utility.

To enter the Alignment utility, when the splash screen has cleared and the normal testing screen appears, press the **Left Button**.

This utility will allow checking and aligning of the laser and receiver pairs by giving an indication on the LCD whether or not the sensor can see the laser beam.

When the laser is shining onto the receiver and the receiver can see the laser, the LCD will report 'Seen' When the laser is blocked, by a piece of card or your hand, the LCD will say 'blocked'.



There are two different versions of the sensor, referred to as 'original' and 'new'. Both look identical, so there is no way to tell by looking, which type they are. The only difference between the two is that one has an inverse output to the other. If the Laser Alignment Utility seems to be working backwards, it means you have the old type of sensor.

To change the sensor type, press the **Right Button**.

To exit the utility, press **Left Button**.

If there is no change at all to the LCD screen, then check all the wiring. Lasers should be shining Red and the LED on the receiver boards should be on. There is no protection from incorrect wiring or sensor orientation on the boards, so incorrect wiring will most likely result in a damaged board or sensor.

Important Note from around 2026 a faulty batch of sensor PCBs were made. On the board, there are two resistors. One reduces the current and feeds the LED. This is fine. However, the 'pull up' resistor, which gives low current feed to the Arduino, is missing a trace on the pcb. This has the effect of making the sensors appear not to change state. Check the pcb to see if your board(s) have this fault.

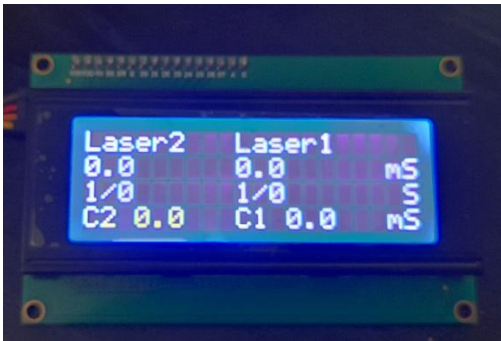
Note: - You will probably find that the Laser barrel is not affixed to the pcb other than by its connecting wires. Adding a bit of hot-glue, or another suitable adhesive, will hold it in place.

To exit the utility, press 'Reset Average' button.

Note: - The Laser Alignment utility is only displayed on the LCD, no output is sent to the PC screen.

Normal Use

After the initial splash screen, the LCD will change to the testing Screen.



Testing screen

There are two columns. Each column will show the calculated exposure measurement from each of the laser and receiver pair.

The measured exposure is shown in milliseconds (0.001 of a second) one decimal place.

The fractional value is then shown below

The last value in each column C2 & C1 show the time Curtain1 and Curtain2 took to travel between the two sensors in milliseconds.

Each time the camera shutter is fired, the display will automatically update.

The letter '**R**' will appear on the top right of the LCD (and 'Ready...' on the PC screen) when the Shutter Tester is ready for a new test. The sensors must be blocked, for this to happen.

If '**B**' appears in the top right corner, it is indicating that shutter bounce has been detected.

Averages.

Pressing the **Left Button** will show the average of (up to) the last ten readings.

Pressing the **Right Button** will reset all the average readings. The last test result is displayed (but not recorded in the averages).

Press the **Left Button** to return to the testing screen.

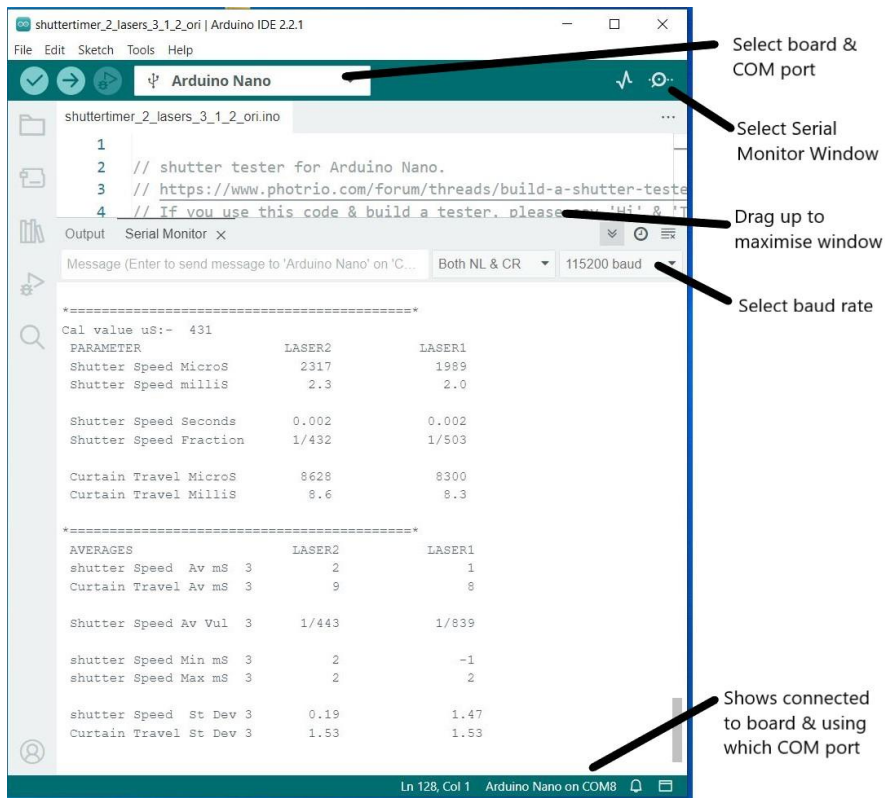
Note: - Due to limitations of the Arduino memory, error checking is limited. So strange readings may occur, for example fingers waved across the lasers or a seriously badly behaving camera is being tested.

LED light, including computer monitors, can also be problematic if it is PWM controlled. The Shutter Tester may see this as sensor changes.

Computer Screen Display

More information can be displayed on the computer screen, than that of the LCD alone.

For this, a computer program called 'Arduino IDE' will be required. See separate document of how to instal this program.



The display follows a similar format to that of the LCD, but with more information.

Shutter bounces will be shown as well as if flash sync has occurred at the given speed.

Averages are updated and displayed. The number of cumulative tests used to calculate the average are shown (in this picture, 3 can be seen).

A maximum of ten individual tests are used to calculate the average. After this, the oldest is lost from the calculation and the latest reading added.

As with the LCD, pressing the 'Reset Average' button will clear the historic data and new averages will be calculated using new test data.

V 2.0.1

Updated wording stating PC screen can also be used for authentication.

V2.0.2

Button functions changed

Button legends appear on the LCD

Ability to change sensor type added.